

Stable Recovery Beyond Spectral Blindness: Finite Sampling, Minimal Observer Lifts, and Completion-Stable Reconstruction

Monty Dabas

July 2026

Abstract

A representation may be exact on the data it consumes and still be blind to target-relevant information discarded at the source. We develop a finite- and Hilbert-space reconstruction theory for restricted parameter-dependent observations, motivated by a companion spectral-blindness model with 50 formal coordinates, collapsed rank 45, and five exact blind modes.

Parameter sampling is treated as a second source restriction on the actual coefficient-response space. After a sample set Q is fixed, the exact minimum number of arbitrary scalar probes required for a declared target Π is

$$\text{rank}(\Pi|_{\ker \mathcal{E}_Q}).$$

Additional samples exchange exactly with the target dimensions they recover, whereas finite probe catalogues obey a prescribed-row-space spanning criterion that need not be a matroid-basis problem. Target blindness also has an exact composition law, giving a nonnegative stage-by-stage quotient ledger.

Uniform visible- and blind-sector margins yield bounded target decoders, deterministic noise control, perturbation stability, and strong completion. For numerically generated response lifts, an a posteriori residual identity bounds the uncaptured seed action by $\|A^{-1/2}E\|$; response-Krylov frontier recursion, metric whitening, and a matched two-block inequality then produce a projection-consistent spectral certificate.

An exact degree-six stress test gives 270 formal necklace coordinates, collapsed rank 209, blind dimension 61, and a minimal 61-probe split-occurrence lift at $q = 2$. Symmetry reduction splits the blind sector into 30 cyclic-reversal-even and 31 cyclic-reversal-odd dimensions, with a fully invariant sector of dimension eight. The results concern declared linear targets and operator representations; they do not assert recovery of erased endpoint history, physical implementation, operator positivity, or any open conjecture.

1 Motivation, lineage, and scope

The immediate parent of this paper is the companion manuscript *When Spectra Forget Order* [2]. That work constructs general multiplicity transforms for ordered matrix products and proves the exact degree-five calibration

$$50 \text{ formal coordinates,} \quad 45 \text{ collapsed functional dimensions,} \quad 5 \text{ blind dimensions.}$$

It also exhibits five split-occurrence probes whose restriction determinant is -2592 , proving that the full five-dimensional defect is exactly and minimally repairable.

The present paper begins where that finite theorem stops. It asks whether the all-parameter blind kernel can be recovered from finitely many measurements, whether fewer measurements can suffice for the actual response space, how many probes are necessary for a particular target rather than the full native state, and when those repairs survive noise and completion.

An earlier thermodynamic-response paper [1] remains a background source for the broader recursive-response and geometric-completion programme. It is not used as the immediate proof source for the 45/50 result or for the theorems proved here.

The paper sits near several established reconstruction frameworks. Generalized sampling studies stable recovery in one reconstruction space from measurements in another and isolates quantitative sampling rates [3, 4]. Frame-completion and erasure theory ask how additional measurement vectors can improve or restore reconstruction [5, 6]. The present emphasis is complementary: it identifies the target-relevant image of a source-restriction kernel, gives its exact arbitrary-scalar lift count, tracks new target blindness under composition, and separates exact identifiability from catalogue achievability and completion-stable margins. No exhaustive priority claim is made; the contribution is the unified blind-kernel calculus and its exact matrix-product calibrations.

The motivating problem is elementary to state and surprisingly persistent: a representation can be internally consistent while failing to preserve the distinction required by a downstream decision. Sampling more values of the same collapsed observable may then reproduce the same blindness with greater numerical confidence.

The present version also tests the theory one degree beyond the parent calibration. The same exact cyclic-descent construction at degree six has 270 formal coordinates, collapsed rank 209, and a 61-dimensional blind kernel. No definition is changed to obtain that classification.

The answer is organized around three objects.

Definition 1.1 (Blind kernel). *For a linear observation $A : X \rightarrow Y$, the blind kernel is*

$$\mathfrak{B}(A) = \ker A.$$

For a parameter family $A(q)$, its all-parameter blind kernel is

$$\mathfrak{B}_{\text{all}}(A) = \bigcap_q \ker A(q),$$

where the parameter set is understood from context.

Definition 1.2 (Target faithfulness). *Let $\Pi : X \rightarrow Z$ be a declared target map. An observation $A : X \rightarrow Y$ is target faithful when*

$$\ker A \subseteq \ker \Pi.$$

Full reconstruction is the special case $\Pi = \text{Id}_X$.

Definition 1.3 (Supplemental lift). *Given $A : X \rightarrow Y$, a supplemental observation bank is a linear map $G : X \rightarrow W$. The augmented observation is*

$$F = (A, G) : X \longrightarrow Y \oplus W.$$

The lift is complete when F is injective, and target complete when $\ker F \subseteq \ker \Pi$.

The core results are finite-dimensional in the parameter-sampling, exact-sequence, and minimum-count sections, and Hilbert-space results in the stable-decoder and completion sections. No claim is made that endpoint-only data recover an event history already erased by the endpoint map. Source-restriction blindness and same-endpoint history blindness are distinct kernels.

2 Finite parameter sampling recovers the all-parameter blind kernel

Let X and Y be finite-dimensional complex vector spaces and let

$$A(q) = \sum_{s=0}^m q^s A_s, \quad A_s \in \mathcal{L}(X, Y).$$

Define the coefficient-stack operator

$$\mathcal{C}_A : X \longrightarrow Y^{m+1}, \quad \mathcal{C}_A x = (A_0 x, \dots, A_m x).$$

For pairwise distinct points

$$Q = (q_0, \dots, q_m),$$

define the sampled observation

$$\mathcal{E}_Q x = (A(q_0)x, \dots, A(q_m)x).$$

Let $V_Q = (q_j^s)_{0 \leq j, s \leq m}$ be the Vandermonde matrix.

Theorem 2.1 (Finite sampling equivalence). *For every $m+1$ pairwise distinct parameter values,*

$$\mathcal{E}_Q = (V_Q \otimes \text{Id}_Y) \mathcal{C}_A.$$

Consequently,

$$\ker \mathcal{E}_Q = \ker \mathcal{C}_A = \bigcap_{s=0}^m \ker A_s = \bigcap_{q \in \mathbb{C}} \ker A(q).$$

Thus $m+1$ distinct parameter samples recover the exact all-parameter blind kernel.

Proof. The j -th component of $(V_Q \otimes \text{Id}_Y) \mathcal{C}_A x$ is

$$\sum_{s=0}^m q_j^s A_s x = A(q_j) x.$$

Since V_Q is invertible, $V_Q \otimes \text{Id}_Y$ is invertible and the first kernel equality follows. The second is the definition of the coefficient stack. Finally, a vector lies in every $\ker A(q)$ exactly when the Y -valued polynomial $q \mapsto A(q)x$ vanishes identically, equivalently when each coefficient $A_s x$ vanishes. $\square$

Theorem 2.2 (Worst-case optimality). *No universal rule using fewer than $m+1$ scalar parameter values can recover the all-parameter blind kernel of every degree- m family.*

Proof. Take $X = \mathbb{C}^{m+1}$, $Y = \mathbb{C}$, and

$$A(q)(x_0, \dots, x_m) = \sum_{s=0}^m x_s q^s.$$

The all-parameter blind kernel is zero. Sampling at $r < m+1$ values produces an $r \times (m+1)$ evaluation matrix, whose kernel is nonzero. $\square$

The theorem has a quantitative version.

Theorem 2.3 (Vandermonde transfer of a lower frame bound). *Assume*

$$\|\mathcal{C}_A x\| \geq c \|x\|, \quad c > 0.$$

Then

$$\|\mathcal{E}_Q x\| \geq \sigma_{\min}(V_Q) c \|x\|.$$

Hence the choice of sampling nodes affects conditioning but not exact identifiability.

Proof. Use

$$\|\mathcal{E}_Q x\| = \|(V_Q \otimes \text{Id}_Y) \mathcal{C}_A x\| \geq \sigma_{\min}(V_Q) \|\mathcal{C}_A x\|.$$

$\square$

Remark 2.4. *The exact kernel theorem requires only distinct points. Stable numerical reconstruction additionally requires a well-conditioned sampling matrix. Repeatedly sampling a poorly conditioned cluster of parameter values is not a substitute for information; it is merely an expensive way to amplify rounding error.*

3 Response-adapted sampling as a second source restriction

The universal $m + 1$ -sample theorem treats every coefficient direction as potentially active. A particular operator family may occupy a much smaller coefficient-response space, so fewer parameter values can suffice. The exact criterion is itself another source-restriction theorem.

Let $\Phi = \text{span}\{\phi_1, \dots, \phi_p\}$ be a finite-dimensional space of scalar functions on a parameter set Ω , and let

$$A(q) = \sum_{s=1}^p \phi_s(q) A_s, \quad A_s \in \mathcal{L}(X, Y),$$

where X and Y are finite-dimensional complex vector spaces. Define the coefficient-stack map

$$\mathcal{C}_\Phi : X \longrightarrow Y^p, \quad \mathcal{C}_\Phi x = (A_1 x, \dots, A_p x),$$

and its coefficient-response space

$$\mathcal{R}_A = \text{ran } \mathcal{C}_\Phi \subseteq Y^p.$$

For a sample set $Q = (q_1, \dots, q_r)$, let

$$E_Q = (\phi_s(q_j))_{1 \leq j \leq r, 1 \leq s \leq p}$$

be the scalar evaluation matrix. The sampled observation is

$$\mathcal{E}_Q x = (A(q_1)x, \dots, A(q_r)x).$$

Theorem 3.1 (Sampling-restriction factorization). *The sampled map factors as*

$$\boxed{\mathcal{E}_Q = (E_Q \otimes \text{Id}_Y) \mathcal{C}_\Phi.}$$

There is a canonical short exact sequence

$$\boxed{0 \longrightarrow \ker \mathcal{C}_\Phi \longrightarrow \ker \mathcal{E}_Q \xrightarrow{\mathcal{C}_\Phi} \mathcal{R}_A \cap \ker(E_Q \otimes \text{Id}_Y) \longrightarrow 0.}$$

Consequently,

$$\boxed{\ker \mathcal{E}_Q = \ker \mathcal{C}_\Phi \iff \mathcal{R}_A \cap \ker(E_Q \otimes \text{Id}_Y) = \{0\}.}$$

Proof. The j -th sampled component is

$$\sum_{s=1}^p \phi_s(q_j) A_s x,$$

which proves the factorization. Apply Theorem 5.1 to

$$X \xrightarrow{\mathcal{C}_\Phi} Y^p \xrightarrow{E_Q \otimes \text{Id}_Y} Y^r.$$

The final equivalence is exactness at the middle term. □

Corollary 3.2 (Universal unisolvent sampling). *If E_Q has full column rank, then*

$$\ker \mathcal{E}_Q = \ker \mathcal{C}_\Phi.$$

In particular, any unisolvent set of p parameter values recovers the exact all-parameter blind kernel for every family with coefficient functions in Φ . The polynomial Vandermonde theorem Theorem 2.1 is the case $\Phi = \text{span}\{1, q, \dots, q^m\}$.

The converse is not necessary for a fixed response family: E_Q may have a large scalar kernel while remaining injective on $\mathcal{R}_A$.

Corollary 3.3 (Information lower bound for response-adapted sampling). *Assume exact recovery of the coefficient-stack kernel and write*

$$d_A = \dim \mathcal{R}_A = \text{rank } \mathcal{C}_\Phi, \quad d_Y = \dim Y.$$

Then every sample set of size r satisfies

$$r \geq \left\lceil \frac{d_A}{d_Y} \right\rceil.$$

This is an information bound; the evaluation constraints may force more samples.

Proof. Exact recovery means that $E_Q \otimes \text{Id}_Y$ is injective on $\mathcal{R}_A$. Its codomain has dimension rd_Y , so $d_A \leq rd_Y$. $\square$

Theorem 3.4 (Two-stage stability factorization). *Let*

$$K = \ker \mathcal{C}_\Phi, \quad c_A = \inf_{0 \neq h \in K^\perp} \frac{\|\mathcal{C}_\Phi h\|}{\|h\|},$$

and define the sampling separation on the actual response space by

$$\eta_Q = \inf_{0 \neq u \in \mathcal{R}_A} \frac{\|(E_Q \otimes \text{Id}_Y)u\|}{\|u\|}.$$

If $c_A > 0$ and $\eta_Q > 0$, then

$$\|\mathcal{E}_Q h\| \geq \eta_Q c_A \|h\| \quad (h \in K^\perp).$$

Thus stable parameter recovery factors into a state-space margin and a sampling-space margin.

Proof. For $h \in K^\perp$, the vector $\mathcal{C}_\Phi h$ lies in $\mathcal{R}_A$, so

$$\|\mathcal{E}_Q h\| \geq \eta_Q \|\mathcal{C}_\Phi h\| \geq \eta_Q c_A \|h\|.$$

$\square$

Corollary 3.5 (Adaptive sample completion). *Let Q be the current sample set and define its surviving coefficient-space blind sector*

$$\mathcal{S}_Q = \mathcal{R}_A \cap \ker(E_Q \otimes \text{Id}_Y).$$

A supplemental sample set R closes all remaining parameter-sampling blindness exactly when

$$\mathcal{S}_Q \cap \ker(E_R \otimes \text{Id}_Y) = \{0\}.$$

Equivalently, the additional parameter values form a supplemental lift on the current sampling-blind sector.

Remark 3.6 (Self-similarity of the blindness problem). *The original source restriction acts on native coordinates. Parameter sampling then acts as a second restriction on the coefficient-response space. Sample design is therefore not external bookkeeping: it is another instance of the same blind-kernel and minimal-lift problem.*

4 Joint sample-and-probe design

Parameter samples and supplemental scalar probes are two ways to reduce the same target-relevant blind sector. This section gives their exact tradeoff.

Let $\mathcal{E}_Q : X \rightarrow Y^{|Q|}$ be the observation generated by a parameter sample set Q , let $\Pi : X \rightarrow Z$ be the target, and define

$$K_Q = \ker \mathcal{E}_Q, \quad b_\Pi(Q) = \mathfrak{B}_\Pi(\mathcal{E}_Q) = \text{rank}(\Pi|_{K_Q}).$$

Theorem 4.1 (Sample-conditioned minimum probe count). *After fixing the parameter sample set Q , the minimum number of arbitrary supplemental scalar observations needed for target faithfulness is exactly*

$$r_{\min}(Q) = b_\Pi(Q).$$

Consequently the exact sample/probe Pareto profile is

$$\{(|Q|, b_\Pi(Q)) : Q \text{ admissible}\}.$$

If one parameter location returns $d_Y = \dim Y$ scalar responses and all scalar channels have equal cost, the minimum total scalar-channel count is

$$\min_Q (d_Y |Q| + b_\Pi(Q)).$$

Proof. Apply the target-relative minimum-lift theorem to the fixed observation $A = \mathcal{E}_Q$. Its target blindness index is $b_\Pi(Q)$, which is therefore both the lower bound and an attainable supplemental scalar count. The remaining statements merely record the resulting tradeoff. $\square$

Definition 4.2 (Target recovery gained by new samples). *For a supplemental parameter set R , define*

$$\rho_\Pi(R | Q) = \dim \frac{\Pi(K_Q)}{\Pi(K_{Q \cup R})}.$$

This is the number of independent target-blind combinations removed by the new parameter samples.

Theorem 4.3 (Sample/probe exchange law). *For every pair of sample sets Q, R ,*

$$b_\Pi(Q \cup R) = b_\Pi(Q) - \rho_\Pi(R | Q).$$

Moreover,

$$0 \leq \rho_\Pi(R | Q) \leq \min\{b_\Pi(Q), |R| \dim Y\}.$$

Thus each target dimension recovered by new samples removes exactly one necessary arbitrary scalar probe, but one sample location can remove at most $\dim Y$ such dimensions.

Proof. Because

$$K_{Q \cup R} = K_Q \cap \ker \mathcal{E}_R,$$

one has $\Pi(K_{Q \cup R}) \subseteq \Pi(K_Q)$. Taking dimensions of the corresponding quotient proves the exchange identity. The quotient dimension cannot exceed $b_\Pi(Q)$. It also cannot exceed the codimension of $K_{Q \cup R}$ inside K_Q , which is at most the rank of $\mathcal{E}_R|_{K_Q}$, hence at most $|R| \dim Y$. $\square$

The preceding theorems allow arbitrary scalar probes. A finite implementation catalogue introduces a genuinely different combinatorial problem.

Let

$$\mathcal{L} = \{\ell_e : e \in E\} \subseteq X^*$$

be a finite family of admissible probes. Restrict them to K_Q , and let

$$T_Q = \text{ran}((\Pi|_{K_Q})^*) \subseteq K_Q^*$$

be the target row space on the current blind sector.

Theorem 4.4 (Fixed-catalogue target criterion). *A subset $J \subseteq E$ is target complete after sampling at Q if and only if*

$$T_Q \subseteq \text{span}\{\ell_e|_{K_Q} : e \in J\}.$$

Therefore every target-complete catalogue subset satisfies

$$|J| \geq b_\Pi(Q),$$

but equality need not be attainable from the declared catalogue.

Proof. Let G_J be the row map formed by the chosen probes. Target completeness is

$$\ker(G_J|_{K_Q}) \subseteq \ker(\Pi|_{K_Q}).$$

For finite-dimensional linear maps, kernel inclusion is equivalent to reverse row-space inclusion, giving the displayed criterion. The cardinality lower bound follows because $\dim T_Q = b_\Pi(Q)$. $\square$

Example 4.5 (Catalogue overhead). *Let $K_Q = \mathbb{C}^2$, let the target be*

$$\Pi(x_1, x_2) = x_1 + x_2,$$

and let the only available probes be

$$\ell_1(x) = x_1, \quad \ell_2(x) = x_2.$$

The target blindness index is one, so one arbitrary linear probe would suffice. Neither catalogue probe alone spans the target row $(1, 1)$, however, and both are required. Hence fixed-catalogue minimum cardinality can exceed the target blindness index.

Corollary 4.6 (Why target-specific catalogue selection is not the full observer matroid). *The full-reconstruction problem asks for a basis of K_Q^* and is a matroid-basis problem. The target-specific fixed-catalogue problem asks for a set whose span contains a prescribed subspace T_Q . Its inclusion-minimal solutions need not have equal cardinality and need not be bases of one matroid. Consequently the greedy minimum-weight basis theorem does not apply in general.*

Remark 4.7 (Lawful optimization statement). *For arbitrary synthesized probes, the exact residual cost after sampling is $b_\Pi(Q)$. For a finite catalogue, the lawful optimization is instead*

$$\min_{Q, J} \left\{ c_{\text{samp}}(Q) + \sum_{e \in J} c_e : T_Q \subseteq \text{span}\{\ell_e|_{K_Q} : e \in J\} \right\}.$$

The target blindness index remains the information lower bound, not an automatic catalogue-achievability theorem.

5 Staged restrictions and the blindness exact sequence

Suppose information is consumed in stages,

$$X \xrightarrow{A} Y \xrightarrow{C} Z.$$

The blind kernel can grow at the second stage even when the first observation is injective on its declared model.

Theorem 5.1 (Blindness short exact sequence). *There is a canonical short exact sequence*

$$0 \longrightarrow \ker A \longrightarrow \ker(CA) \xrightarrow{\phi} \ker(C|_{\text{ran } A}) \longrightarrow 0,$$

where

$$\phi(x) = Ax.$$

Consequently, in finite dimensions,

$$\dim \ker(CA) = \dim \ker A + \dim(\ker C \cap \text{ran } A).$$

Proof. If $x \in \ker(CA)$, then $Ax \in \text{ran } A$ and $C(Ax) = 0$, so ϕ is well defined. Its kernel is exactly $\ker A$. It is surjective because every $y \in \ker(C|_{\text{ran } A})$ has the form $y = Ax$ for some x , and then $x \in \ker(CA)$. Exactness and the dimension formula follow. $\square$

Definition 5.2 (Blindness increment). *For a second-stage restriction C acting after A , define*

$$\Delta\mathfrak{B}(C | A) = \dim(\ker C \cap \text{ran } A).$$

This is the new blindness created by C on information that actually survived the first stage.

Corollary 5.3 (Additive blindness ledger). *Let*

$$T_j = C_j C_{j-1} \cdots C_1 A_0$$

for $j \geq 1$, with $T_0 = A_0$. Then

$$\dim \ker T_r = \dim \ker A_0 + \sum_{j=1}^r \dim(\ker C_j \cap \text{ran } T_{j-1}).$$

Proof. Apply Theorem 5.1 recursively to $T_j = C_j T_{j-1}$. $\square$

Remark 5.4 (Why this is more informative than rank alone). *The formula assigns every lost dimension to the stage that destroys it. A final rank defect can therefore be decomposed into contributions from sampling, parity restriction, source-row deletion, quotienting, truncation, or terminal decision compression. Without this ledger, unrelated losses are easily blamed on the last map in the chain.*

Corollary 5.5 (Target-relative staged faithfulness). *Let $\Pi : X \rightarrow W$ be a target. The composition CA is target faithful if and only if*

$$\ker(CA) \subseteq \ker \Pi.$$

Equivalently, both the inherited kernel $\ker A$ and every lifted class in $\ker(C|_{\text{ran } A})$ must be target irrelevant after pullback to X .

6 Composition laws for target blindness

The ordinary blindness exact sequence records how the full kernel grows under composition. For a declared target, only the image of that kernel under the target map is relevant. This yields a second, target-valued composition law.

Let

$$X \xrightarrow{A} Y \xrightarrow{C} Z, \quad \Pi : X \rightarrow W,$$

and write

$$K_A = \ker A, \quad K_{CA} = \ker(CA).$$

Since $K_A \subseteq K_{CA}$, one has $\Pi(K_A) \subseteq \Pi(K_{CA})$.

Definition 6.1 (Target blindness index). *The target blindness index of A relative to Π is*

$$\mathfrak{B}_\Pi(A) = \dim \Pi(\ker A) = \text{rank}(\Pi|_{\ker A}).$$

Definition 6.2 (Target blindness increment). *The new target-relevant blindness created by C after A is*

$$\Delta_\Pi(C \mid A) = \dim \frac{\Pi(K_{CA})}{\Pi(K_A)}.$$

Theorem 6.3 (Target-blindness composition law). *For every composable pair A, C and every linear target Π ,*

$$\mathfrak{B}_\Pi(CA) = \mathfrak{B}_\Pi(A) + \Delta_\Pi(C \mid A).$$

Moreover, if

$$\theta : K_{CA}/K_A \longrightarrow \ker(C|_{\text{ran } A}), \quad \theta([x]) = Ax,$$

is the canonical isomorphism from Theorem 5.1, then the increment equals the rank of the quotient target map

$$\bar{\Pi}_A : K_{CA}/K_A \longrightarrow W/\Pi(K_A), \quad \bar{\Pi}_A([x]) = \Pi x + \Pi(K_A).$$

Thus

$$\Delta_\Pi(C \mid A) = \text{rank } \bar{\Pi}_A.$$

Proof. The inclusions of target images give a short exact sequence

$$0 \longrightarrow \Pi(K_A) \longrightarrow \Pi(K_{CA}) \longrightarrow \Pi(K_{CA})/\Pi(K_A) \longrightarrow 0.$$

Taking dimensions proves the displayed composition law.

The map $\bar{\Pi}_A$ is well defined because replacing x by $x + k$, with $k \in K_A$, changes Πx by an element of $\Pi(K_A)$. Its image is exactly $\Pi(K_{CA})/\Pi(K_A)$, so its rank is the target blindness increment. $\square$

Corollary 6.4 (Target-blindness ledger for a chain). *Let*

$$T_0 = A_0, \quad T_j = C_j T_{j-1} \quad (1 \leq j \leq r).$$

Then

$$\mathfrak{B}_\Pi(T_r) = \mathfrak{B}_\Pi(A_0) + \sum_{j=1}^r \Delta_\Pi(C_j \mid T_{j-1}).$$

In particular, T_r is target faithful exactly when every term in this nonnegative ledger vanishes.

Proof. Apply Theorem 6.3 recursively. Every increment is a quotient dimension and hence nonnegative. The final target blindness index vanishes exactly when $\Pi(\ker T_r) = \{0\}$, which is equivalent to target faithfulness. $\square$

Corollary 6.5 (Direct-sum target law). *Suppose*

$$A = A_1 \oplus A_2, \quad \Pi = \Pi_1 \oplus \Pi_2,$$

with separate target codomains. Then

$$\mathfrak{B}_\Pi(A) = \mathfrak{B}_{\Pi_1}(A_1) + \mathfrak{B}_{\Pi_2}(A_2).$$

If the two target maps instead take values in one common codomain, then

$$\mathfrak{B}_\Pi(A) = \dim(\Pi_1(\ker A_1) + \Pi_2(\ker A_2)),$$

so overlap between target images prevents naive additivity.

Remark 6.6 (What the increment measures). *The ordinary increment $\dim(\ker C \cap \text{ran } A)$ counts every newly erased state direction. The target increment counts only the quotient of those newly erased directions that creates a new target variation beyond the target variation already hidden by A . It therefore avoids charging the same terminal ambiguity twice.*

7 Target-relative minimal lifts and stable decoders

Full reconstruction may be unnecessary. A downstream target can ignore some blind directions, and those directions should not be counted as mandatory observer channels.

Let X, Y, Z be finite-dimensional complex vector spaces, let

$$A : X \rightarrow Y, \quad \Pi : X \rightarrow Z,$$

and put

$$K = \ker A, \quad N = K \cap \ker \Pi.$$

The quotient K/N is the target-relevant part of the blind kernel.

Definition 7.1 (Target blindness index). *Define*

$$\mathfrak{B}_\Pi(A) = \text{rank}(\Pi|_K) = \dim K - \dim N.$$

Theorem 7.2 (Exact target-relative minimum lift count). *Among all scalar banks*

$$G : X \rightarrow \mathbb{C}^r$$

for which the augmented observation is target faithful,

$$\ker(A, G) \subseteq \ker \Pi,$$

the minimum possible number of scalar channels is

$$r_{\min} = \mathfrak{B}_\Pi(A) = \text{rank}(\Pi|_{\ker A}).$$

At the minimum, the restriction of G to K has kernel exactly N and induces an isomorphism

$$K/N \cong \mathbb{C}^{r_{\min}}.$$

Proof. Target faithfulness is equivalent to

$$K \cap \ker G \subseteq N.$$

Hence

$$\ker(G|_K) \subseteq N,$$

and rank-nullity gives

$$\text{rank}(G|_K) = \dim K - \dim \ker(G|_K) \geq \dim K - \dim N = \mathfrak{B}_\Pi(A).$$

Since $\text{rank}(G|_K) \leq r$, every lawful bank has $r \geq \mathfrak{B}_\Pi(A)$.

For attainment, choose a direct sum

$$K = N \oplus Q.$$

Then $\Pi|_Q$ is injective and $\dim Q = \mathfrak{B}_\Pi(A)$. Choose an isomorphism

$$T : \Pi(Q) \longrightarrow \mathbb{C}^{\mathfrak{B}_\Pi(A)}$$

and define on K

$$G(n + q) = T\Pi q.$$

Extend G linearly to X . Its kernel on K is exactly N , so the augmented map is target faithful and attains the lower bound. $\square$

Corollary 7.3 (Full reconstruction as a special case). *For $\Pi = \text{Id}_X$,*

$$\mathfrak{B}_{\text{Id}}(A) = \dim \ker A,$$

so Theorem 7.2 reduces to the full minimal-lift theorem.

We now pass from exact target faithfulness to stable target recovery.

Definition 7.4 (Target stability modulus). *For bounded Hilbert-space operators*

$$F : X \rightarrow W, \quad \Pi : X \rightarrow Z,$$

with $\Pi \neq 0$, define

$$\delta_\Pi(F) = \inf_{\Pi x \neq 0} \frac{\|Fx\|}{\|\Pi x\|}.$$

Theorem 7.5 (Target factorization and optimal decoder). *For bounded operators between Hilbert spaces, the following are equivalent:*

- (i) $\delta_\Pi(F) > 0$;
- (ii) *there exists a bounded decoder $D : W \rightarrow Z$ such that*

$$\Pi = DF;$$

- (iii) *there exists $c < \infty$ such that*

$$\Pi^* \Pi \leq c^2 F^* F.$$

Moreover,

$$\inf\{\|D\| : \Pi = DF\} = \delta_\Pi(F)^{-1}.$$

In finite dimensions, these conditions are equivalent to

$$\ker F \subseteq \ker \Pi.$$

Proof. If $\delta_\Pi(F) > 0$, define on $\text{ran } F$

$$D_0(Fx) = \Pi x.$$

The definition is lawful because $Fx = 0$ implies $\Pi x = 0$, and

$$\|D_0(Fx)\| \leq \delta_\Pi(F)^{-1} \|Fx\|.$$

Extend D_0 continuously to $\overline{\text{ran } F}$ and set it to zero on the orthogonal complement. This gives $\Pi = DF$ and $\|D\| \leq \delta_\Pi(F)^{-1}$.

Conversely, $\Pi = DF$ implies

$$\|\Pi x\| \leq \|D\| \|Fx\|,$$

so $\delta_\Pi(F) \geq \|D\|^{-1}$. Taking the infimum over decoders proves the norm identity. The equivalence with the operator inequality is the Douglas factorization theorem [8]. In finite dimensions, the kernel condition makes the induced decoder on $\text{ran } F$ automatically bounded. $\square$

Let X, Y, W, Z be Hilbert spaces, let $A \in \mathcal{L}(X, Y)$, and set

$$K = \ker A, \quad H = K^\perp, \quad N = K \cap \ker \Pi, \quad Q = K \ominus N.$$

Assume

$$\begin{aligned} \|Ah\| &\geq \sigma \|h\| \quad (h \in H), \\ GN &= \{0\}, \quad \|Gq\| \geq \gamma \|q\| \quad (q \in Q), \end{aligned}$$

and write

$$L = \|G|_H\|, \quad p_H = \|\Pi|_H\|, \quad p_Q = \|\Pi|_Q\|.$$

Theorem 7.6 (Explicit robust target lift). *Define*

$$C_\Pi = \left[\left(\frac{p_H}{\sigma} + \frac{p_Q L}{\gamma \sigma} \right)^2 + \left(\frac{p_Q}{\gamma} \right)^2 \right]^{1/2}.$$

Then $F = (A, G)$ satisfies

$$\boxed{\|\Pi x\| \leq C_\Pi \|Fx\| \quad (x \in X).}$$

Consequently,

$$\delta_\Pi(F) \geq C_\Pi^{-1},$$

and noisy observations of size ε yield target error at most $C_\Pi \varepsilon$ under the canonical bounded decoder.

Proof. Write $x = h + q + n$ with $h \in H$, $q \in Q$, and $n \in N$. Put

$$y = Ah, \quad z = Gq + Gh.$$

Since $\Pi n = 0$,

$$\|\Pi x\| \leq p_H \|h\| + p_Q \|q\|.$$

The visible and target-relevant blind-sector bounds give

$$\begin{aligned} \|h\| &\leq \sigma^{-1} \|y\|, \\ \|q\| &\leq \gamma^{-1} \|z\| + L(\gamma\sigma)^{-1} \|y\|. \end{aligned}$$

Therefore

$$\|\Pi x\| \leq \left(\frac{p_H}{\sigma} + \frac{p_Q L}{\gamma \sigma} \right) \|y\| + \frac{p_Q}{\gamma} \|z\|.$$

Cauchy–Schwarz in $\mathbb{R}^2$ gives the claimed bound. $\square$

Remark 7.7 (What the count removes). *Full repair counts every direction in K . Target-relative repair counts only the quotient $K/(K \cap \ker \Pi)$. A target-null blind direction may remain invisible without weakening the declared conclusion.*

8 Equivariant target reduction

Target blindness can contract sharply when both the observation and the target respect a symmetry. The point is not that symmetry removes native information; it identifies which irreducible blind sectors a declared target is capable of seeing.

Let a finite group Γ act unitarily on finite-dimensional complex spaces X , Y , and Z , with representations ρ_X , ρ_Y , and ρ_Z . Suppose

$$A\rho_X(g) = \rho_Y(g)A, \quad \Pi\rho_X(g) = \rho_Z(g)\Pi$$

for every $g \in \Gamma$.

Theorem 8.1 (Equivariant target-blindness decomposition). *The blind kernel $K = \ker A$ is a Γ -subrepresentation. Writing its isotypic decomposition as*

$$K = \bigoplus_{\tau \in \widehat{\Gamma}} K[\tau],$$

one has

$$\Pi(K[\tau]) \subseteq Z[\tau]$$

and therefore

$$\boxed{\mathfrak{B}_\Pi(A) = \sum_{\tau \in \widehat{\Gamma}} \text{rank}(\Pi|_{K[\tau]}).$$

If the irreducible type τ does not occur in Z , then $K[\tau] \subseteq \ker \Pi$.

Proof. Equivariance gives $A\rho_X(g)K = 0$, so K is invariant. Finite-group representations over $\mathbb{C}$ are semisimple. The restriction of an equivariant map to a τ -isotypic component has image in the τ -isotypic component of the codomain. The image $\Pi(K)$ is the direct sum of these isotypic images, so its dimension is the displayed sum. If $Z[\tau] = 0$, the corresponding image vanishes. $\square$

Corollary 8.2 (Invariant target). *If Γ acts trivially on Z , then*

$$\Pi(K) = \Pi(K^\Gamma), \quad \mathfrak{B}_\Pi(A) \leq \dim K^\Gamma.$$

Thus every nontrivial isotypic blind sector is target-null for an invariant target.

Corollary 8.3 (Central involution). *Let J be a central involution commuting with A , and write*

$$K = K^+ \oplus K^-, \quad K^\pm = \ker(J \mp I) \cap K.$$

If $\Pi J = \Pi$, then $K^- \subseteq \ker \Pi$ and

$$\mathfrak{B}_\Pi(A) = \text{rank}(\Pi|_{K^+}) \leq \dim K^+.$$

If $\Pi J = -\Pi$, the analogous statement holds with K^- .

Proof. For $x \in K^-$, invariance gives $\Pi x = \Pi J x = -\Pi x$, hence $\Pi x = 0$. The odd-target case is identical. $\square$

The same symmetry may be imposed on the lift bank. Let $\mathcal{G} \subset K^*$ be a complete finite probe family stable under the contragredient group action. For a central involution J , define

$$g^\pm = \frac{1}{2}(g \pm g \circ J).$$

Proposition 8.4 (Symmetry-adapted lift bank). *If $\mathcal{G}$ spans K^* , then the projected families*

$$\mathcal{G}^+ = \{g^+ : g \in \mathcal{G}\}, \quad \mathcal{G}^- = \{g^- : g \in \mathcal{G}\}$$

span $(K^+)^$ and $(K^-)^*$, respectively. Hence there exist complete symmetry-adapted subbanks of cardinalities*

$$\dim K^+, \quad \dim K^-.$$

Proof. The dual idempotents $(I \pm J^*)/2$ project K^* onto the dual even and odd subspaces. Applying a surjective projection to a spanning family gives a spanning family of the image. Extract bases from those projected spans. $\square$

This result is target-relative rather than cosmetic. A symmetry-invariant terminal quantity should be charged only for invariant blind modes. Conversely, a symmetry-breaking target must declare the additional isotypic sectors it consumes; silently discarding them is another source restriction.

9 Robust minimal lifts

Let X, Y, Z be Hilbert spaces and let $A \in \mathcal{L}(X, Y)$. Put

$$K = \ker A, \quad H = K^\perp.$$

Assume that the visible observation is quantitatively faithful on H :

$$\|Ah\| \geq \sigma\|h\|, \quad h \in H, \quad \sigma > 0.$$

Let $G \in \mathcal{L}(X, Z)$ be a supplemental bank satisfying

$$\|Gk\| \geq \gamma\|k\|, \quad k \in K, \quad \gamma > 0.$$

Write

$$L = \|G|_H\|.$$

Theorem 9.1 (Quantitative blind-kernel lift). *Define*

$$T_{\sigma, \gamma, L} = \begin{pmatrix} \sigma^{-1} & 0 \\ L(\gamma\sigma)^{-1} & \gamma^{-1} \end{pmatrix}$$

and

$$\delta(\sigma, \gamma, L) = \|T_{\sigma, \gamma, L}\|^{-1}.$$

Then the augmented observation

$$F = (A, G) : X \rightarrow Y \oplus Z$$

satisfies

$$\boxed{\|Fx\| \geq \delta(\sigma, \gamma, L)\|x\| \quad (x \in X).}$$

In particular, F is injective, has closed range, and its inverse on $\text{ran } F$ has norm at most δ^{-1} .

Proof. Write $x = k + h$ with $k \in K$, $h \in H$, and set

$$y = Ah, \quad z = Gk + Gh.$$

The visible lower bound gives

$$\|h\| \leq \sigma^{-1}\|y\|.$$

The blind-sector lower bound gives

$$\|k\| \leq \gamma^{-1} \|Gk\| \leq \gamma^{-1} \|z\| + L(\gamma\sigma)^{-1} \|y\|.$$

Therefore

$$\left(\frac{\|h\|}{\|k\|} \right) \leq T_{\sigma, \gamma, L} \left(\frac{\|y\|}{\|z\|} \right)$$

componentwise. Taking Euclidean norms yields

$$\|x\| = \sqrt{\|h\|^2 + \|k\|^2} \leq \|T_{\sigma, \gamma, L}\| \sqrt{\|y\|^2 + \|z\|^2} = \delta^{-1} \|Fx\|.$$

□

Corollary 9.2 (Orthogonal lift). *If $G(H) = \{0\}$, then $L = 0$ and*

$$\boxed{\|Fx\| \geq \min\{\sigma, \gamma\} \|x\|}.$$

Corollary 9.3 (Noise-stable reconstruction). *Let*

$$w = Fx + \eta, \quad \|\eta\| \leq \varepsilon,$$

and let $\hat{x} = F^\dagger w$ be the minimum-norm reconstruction. Then

$$\boxed{\|\hat{x} - x\| \leq \frac{\varepsilon}{\delta(\sigma, \gamma, L)}}.$$

Proof. Since F is bounded below by δ , its Moore–Penrose inverse on $\text{ran } F$ has norm at most δ^{-1} . □

Theorem 9.4 (Perturbation stability). *If $\tilde{F} \in \mathcal{L}(X, Y \oplus Z)$ satisfies*

$$\|\tilde{F} - F\| < \delta,$$

then

$$\boxed{\|\tilde{F}x\| \geq (\delta - \|\tilde{F} - F\|) \|x\|}.$$

Thus completeness and stable reconstruction survive every perturbation smaller than the certified lower margin.

Proof. Apply the reverse triangle inequality:

$$\|\tilde{F}x\| \geq \|Fx\| - \|(\tilde{F} - F)x\|.$$

□

Remark 9.5 (Minimality versus robustness). *The dimension bound*

$$\dim Z \geq \dim K$$

is necessary for full finite-dimensional repair, but it says nothing about conditioning. A square bank may be injective on K while having an arbitrarily small γ . Minimality counts probes; robustness measures the weakest surviving blind direction.

10 Response-adapted lifts and a posteriori residual certificates

Exact lift counts determine how many target-relevant blind directions must be repaired. In computation, however, the lift vectors themselves are commonly obtained from linear solves and are therefore only approximate. This section turns the solve residual into a rigorous bound on the unresolved response leakage. Residual-based spectral approximation is classical [7]; the point here is to connect such residuals to a source-space lift that is constructed from operator responses and to enforce one matched projection throughout the final block certificate.

Let X be a finite-dimensional Hilbert space and let $A > 0$ define the metric

$$\langle x, y \rangle_A = \langle Ax, y \rangle.$$

Let $T_0, T_1 \in \mathcal{L}(X)$ be self-adjoint in this metric. Equivalently,

$$AT_j = T_j^* A, \quad j = 0, 1.$$

Write

$$B_1 = AT_1,$$

so that $B_1 = B_1^*$ in the original Hilbert structure.

Let $U_0 \subseteq X$ be a finite-dimensional seed space satisfying

$$\text{ran } T_0 \subseteq U_0.$$

Choose an A -orthonormal synthesis map

$$V_0 : \mathbb{C}^r \longrightarrow X, \quad V_0^* A V_0 = I,$$

and let

$$P_0 = V_0 V_0^* A$$

be the A -orthogonal projection onto U_0 .

For the exact response columns one would solve

$$AR = B_1 V_0, \quad R = T_1 V_0.$$

Suppose instead that a numerical procedure returns $\tilde{R}$, with residual

$$E = A\tilde{R} - B_1 V_0.$$

Define the enriched space and its matched projections by

$$\tilde{U}_1 = U_0 + \text{ran } \tilde{R}, \quad P_1 = P_{\tilde{U}_1}^{(A)}, \quad Q_1 = I - P_1.$$

Finally put

$$\rho = \|A^{-1/2} E\|_{\mathbb{C}^r \rightarrow X}.$$

Theorem 10.1 (A posteriori response-defect identity). *The uncaptured action of T_1 on the seed space satisfies*

$$Q_1 T_1 P_0 = -Q_1 A^{-1} E V_0^* A.$$

Consequently,

$$\|Q_1 T_1 P_0\|_A \leq \rho.$$

Proof. The residual equation gives

$$T_1 V_0 = A^{-1} B_1 V_0 = \tilde{R} - A^{-1} E.$$

Since $\text{ran } \tilde{R} \subseteq \tilde{U}_1$, one has $Q_1 \tilde{R} = 0$. Multiplying by $V_0^* A$ proves the identity. In the A -metric, Q_1 is an orthogonal projection and $V_0^* A$ is a coisometry. Hence

$$\|Q_1 A^{-1} E V_0^* A\|_A \leq \|A^{-1/2} E\|.$$

□

Corollary 10.2 (Frontier plus residual leakage). *Let $\tilde{T} = T_0 + T_1$. Then*

$$Q_1 \tilde{T} P_1 = Q_1 T_1 (P_1 - P_0) - Q_1 A^{-1} E V_0^* A.$$

Thus, with

$$\beta_1 = \|Q_1 T_1 (P_1 - P_0)\|_A,$$

one has

$$\|Q_1 \tilde{T} P_1\|_A \leq \beta_1 + \rho.$$

Proof. The range hypothesis gives $Q_1 T_0 = 0$. Split $P_1 = P_0 + (P_1 - P_0)$ and apply Theorem 10.1 to the seed term. □

The exact case has a useful recursive form. Assume now that all response columns are exact and define

$$U_m = U_0 + T_1 U_0 + \cdots + T_1^m U_0, \quad m \geq 1.$$

Let P_m be the A -orthogonal projection onto U_m , with $Q_m = I - P_m$.

Proposition 10.3 (Response–Krylov frontier recursion). *For every $m \geq 1$,*

$$Q_m (T_0 + T_1) P_{m-1} = 0.$$

Consequently,

$$Q_m (T_0 + T_1) P_m = Q_m T_1 (P_m - P_{m-1}).$$

Only the newest response frontier can leak from the stage- m space.

Proof. The range of T_0 lies in $U_0 \subseteq U_m$. Moreover,

$$T_1 U_{m-1} \subseteq T_1 U_0 + \cdots + T_1^m U_0 \subseteq U_m.$$

This proves the first identity. The second follows by decomposing $P_m = P_{m-1} + (P_m - P_{m-1})$. □

The finite eigenvalue calculation must use the same enriched space. Let

$$\Phi_1 : \mathbb{C}^d \rightarrow X$$

be any injective synthesis map with range $\tilde{U}_1$, and define

$$H_1 = \Phi_1^* A \Phi_1, \quad C_1 = \Phi_1^* A \tilde{T} \Phi_1.$$

Proposition 10.4 (Metric-whitened response compression). *The compression of $\tilde{T}$ to the matched response space is represented in orthonormal coordinates by*

$$K_1 = H_1^{-1/2} C_1 H_1^{-1/2}.$$

Its spectrum is invariant under every invertible change of synthesis coordinates.

Proof. The synthesis map $\Phi_1 H_1^{-1/2}$ is A -orthonormal. Compressing $\tilde{T}$ through that map gives the displayed matrix. A change $\Phi_1 \mapsto \Phi_1 S$ changes the generalized eigenvalue pencil by congruence and leaves its spectrum unchanged. $\square$

Define the matched block quantities

$$a_1 = \sup \sigma(P_1 \tilde{T} P_1|_{\tilde{U}_1}), \quad b_1 = \|Q_1 \tilde{T} P_1\|_A, \quad d_1 = \sup \sigma(Q_1 \tilde{T} Q_1|_{Q_1 X}).$$

Theorem 10.5 (Matched two-block extremal certificate). *One has*

$$\sup \sigma(\tilde{T}) \leq \frac{a_1 + d_1 + \sqrt{(a_1 - d_1)^2 + 4b_1^2}}{2}.$$

If directed computation supplies upper bounds

$$a_1 \leq \bar{a}_1, \quad \beta_1 + \rho \leq \bar{b}_1, \quad d_1 \leq \bar{d}_1,$$

then the same formula with bars gives a certified upper bound on $\sup \sigma(\tilde{T})$.

Proof. For a unit vector $x = u + v$, with $u = P_1 x$ and $v = Q_1 x$,

$$\langle \tilde{T} x, x \rangle_A \leq a_1 \|u\|_A^2 + 2b_1 \|u\|_A \|v\|_A + d_1 \|v\|_A^2.$$

The maximum of the scalar quadratic form is the largest eigenvalue of

$$\begin{pmatrix} a_1 & b_1 \\ b_1 & d_1 \end{pmatrix}.$$

The residual bound follows from Theorem 10.2 and monotonicity of the displayed expression in $b_1 \geq 0$. $\square$

Corollary 10.6 (Transfer from an approximate operator). *Let T be another A -self-adjoint operator satisfying*

$$\|T - \tilde{T}\|_A \leq \varepsilon_{\text{tail}}.$$

Then

$$\sup \sigma(T) \leq \frac{\bar{a}_1 + \bar{d}_1 + \sqrt{(\bar{a}_1 - \bar{d}_1)^2 + 4\bar{b}_1^2}}{2} + \varepsilon_{\text{tail}}.$$

Every term must refer to the same metric A and the same projection P_1 .

Remark 10.7 (Information boundary). *A residual certificate repairs approximate response vectors; it does not remove the complement obligation d_1 . A small solve residual and a favourable finite compression cannot exclude a crossing wholly inside $Q_1 X$. The finite block, cross action, complement bound, and operator tail are one matched certificate, not interchangeable estimates from unrelated approximations.*

11 Completion-stable blind-kernel lifts

The previous theorem is finite-stage. We now ask when one family of repaired observations remains repaired in the limit.

Let X, Y, Z be Hilbert spaces. For each n , let P_n be an orthogonal projection on X , and let

$$A_n \in \mathcal{L}(X, Y), \quad G_n \in \mathcal{L}(X, Z).$$

Think of $P_n X$ as the stage- n blind sector.

Theorem 11.1 (Strong completion of uniformly repaired observations). *Assume:*

- (i) $P_n \rightarrow P$ strongly;
- (ii) $A_n \rightarrow A$ and $G_n \rightarrow G$ strongly;
- (iii) $A_n P_n = 0$;
- (iv) there exists $\sigma > 0$ such that

$$\|A_n h\| \geq \sigma \|h\|, \quad h \in (I - P_n)X;$$

- (v) there exists $\gamma > 0$ such that

$$\|G_n k\| \geq \gamma \|k\|, \quad k \in P_n X;$$

- (vi) there exists $L < \infty$ such that

$$\|G_n(I - P_n)\| \leq L.$$

Then:

$$\begin{aligned} AP &= 0, & \ker A &= PX, \\ \|Ah\| &\geq \sigma \|h\|, & h &\in (I - P)X, \\ \|Gk\| &\geq \gamma \|k\|, & k &\in PX, \end{aligned}$$

and

$$\|G(I - P)\| \leq L.$$

Moreover, with

$$F_n = (A_n, G_n), \quad F = (A, G),$$

the common constant

$$\delta = \delta(\sigma, \gamma, L)$$

from Theorem 9.1 satisfies

$$\boxed{\|F_n x\| \geq \delta \|x\| \quad \text{for every } n,}$$

and

$$\boxed{\|F x\| \geq \delta \|x\|}.$$

Thus the blind-kernel repair survives strong completion with no loss of the certified lower margin.

Proof. Strong convergence implies uniform boundedness of (A_n) and (G_n) . For $x \in X$,

$$A_n P_n x = 0$$

and $P_n x \rightarrow Px$; hence $APx = 0$.

If $h \in (I - P)X$, put $h_n = (I - P_n)h$. Then $h_n \rightarrow h$, and

$$A_n h_n \rightarrow Ah.$$

Therefore

$$\|Ah\| = \lim_n \|A_n h_n\| \geq \sigma \lim_n \|h_n\| = \sigma \|h\|.$$

This proves $\ker A = PX$.

For $k \in PX$, put $k_n = P_n k$. Then $k_n \rightarrow k$ and

$$G_n k_n \rightarrow Gk,$$

so

$$\|Gk\| \geq \gamma \|k\|.$$

The cross bound follows similarly from $h_n = (I - P_n)h$ for $h \in (I - P)X$.

Each F_n satisfies the hypotheses of Theorem 9.1 with the same three constants, so $\|F_n x\| \geq \delta \|x\|$. Since $F_n x \rightarrow Fx$ for each fixed x , the limit inequality follows. $\square$

Corollary 11.2 (Transported varying-space version). *Let X_n, Y_n, Z_n be Hilbert spaces connected to fixed model spaces X, Y, Z by unitary identifications*

$$U_n : X \rightarrow X_n, \quad V_n : Y \rightarrow Y_n, \quad W_n : Z \rightarrow Z_n.$$

If the transported operators

$$\hat{A}_n = V_n^* A_n U_n, \quad \hat{G}_n = W_n^* G_n U_n, \quad \hat{P}_n = U_n^* P_n U_n$$

satisfy the hypotheses of Theorem 11.1, then the native augmented observations (A_n, G_n) have the same uniform lower bound and their transported limit is completion-stable.

Example 11.3 (Why stagewise injectivity is not enough). *Let $A_n = (1/n)\text{Id}$ and $G_n = 0$. Every A_n is injective, but the lower margin tends to zero and the strong limit is the zero operator. The theorem fails exactly where it should: no uniform σ exists.*

Remark 11.4. *The uniform constants σ and γ are mathematical data, not decorative tolerances. A finite lift whose weakest blind-sector singular value collapses under refinement does not survive completion.*

11.1 Target-relative completion

Full injectivity is stronger than many applications require. The next theorem allows a target-null part of the blind sector to persist through refinement.

Theorem 11.5 (Completion of uniformly target-repaired observations). *Let X, Y, W, Z be Hilbert spaces. For each n , let P_n and N_n be orthogonal projections with*

$$N_n X \subseteq P_n X,$$

and put

$$Q_n = P_n - N_n, \quad H_n = I - P_n.$$

Assume

$$P_n \rightarrow P, \quad N_n \rightarrow N$$

strongly, and let

$$A_n \rightarrow A, \quad G_n \rightarrow G, \quad \Pi_n \rightarrow \Pi$$

strongly. Suppose there are constants

$$\sigma, \gamma > 0, \quad L, p_H, p_Q < \infty$$

independent of n such that

$$A_n P_n = 0, \quad \|A_n h\| \geq \sigma \|h\| \quad (h \in H_n X),$$

$$G_n N_n = 0, \quad \|G_n q\| \geq \gamma \|q\| \quad (q \in Q_n X), \quad \|G_n H_n\| \leq L,$$

and

$$\Pi_n N_n = 0, \quad \|\Pi_n H_n\| \leq p_H, \quad \|\Pi_n Q_n\| \leq p_Q.$$

Define

$$C = \left[\left(\frac{p_H}{\sigma} + \frac{p_Q L}{\gamma \sigma} \right)^2 + \left(\frac{p_Q}{\gamma} \right)^2 \right]^{1/2}.$$

Then every stage satisfies

$$\boxed{\|\Pi_n x\| \leq C \|(A_n x, G_n x)\| \quad (x \in X).}$$

Moreover, $N \leq P$, with

$$Q = P - N, \quad H = I - P,$$

and the limit operators satisfy

$$\|\Pi x\| \leq C\|(Ax, Gx)\| \quad (x \in X).$$

Thus target reconstruction survives strong completion even though the permanent target-null blind sector NX remains unreconstructed.

Proof. Since $N_n \leq P_n$, one has

$$P_n N_n = N_n P_n = N_n.$$

Strong convergence gives $PN = NP = N$, so $N \leq P$, and $Q_n \rightarrow Q = P - N$, $H_n \rightarrow H = I - P$ strongly.

At each finite stage, decompose

$$x = H_n x + Q_n x + N_n x.$$

The hypotheses are exactly those of Theorem 7.6, with uniform constants, and therefore give the stagewise inequality.

The same approximation argument used in Theorem 11.1 passes every structural estimate to the limit. For example, if $h \in HX$, then $H_n h \rightarrow h$ and

$$A_n H_n h \rightarrow Ah,$$

so $\|Ah\| \geq \sigma\|h\|$. Likewise, for $q \in QX$, $Q_n q \rightarrow q$ and $\|Gq\| \geq \gamma\|q\|$. The annihilation and norm bounds pass to the limit in the same way. Apply Theorem 7.6 to (A, G, Π) . $\square$

Remark 11.6. *The theorem distinguishes a lawful persistent kernel from an uncontrolled collapse. Target-null blindness may survive completion; target-relevant blindness must retain the uniform lift margin γ .*

12 Observer matroids and minimum-cost complete banks

Let K be a d -dimensional blind space and let

$$\mathcal{G} = \{g_e : e \in E\} \subseteq K^*$$

be a finite family of admissible scalar probes. For the full-reconstruction statements below, assume that the catalogue is complete in principle:

$$\text{span } \mathcal{G} = K^*.$$

Theorem 12.1 (Observer matroid). *Define*

$$\mathcal{I} = \{J \subseteq E : \{g_e : e \in J\} \text{ is linearly independent in } K^*\}.$$

Then $(E, \mathcal{I})$ is a representable matroid of rank d . A subset $J \subseteq E$ is an inclusion-minimal complete observer bank on K if and only if it is a matroid basis. Consequently every minimal complete bank has exactly d probes.

Proof. The independent subsets of a finite vector configuration satisfy the matroid axioms. The spanning assumption makes the matroid rank equal to d . A bank is complete on K exactly when its restrictions span K^* , and it is inclusion-minimal complete exactly when those restrictions form a basis of K^* . These are precisely the matroid bases. $\square$

Corollary 12.2 (Minimum-cost exact repair). *Assign a nonnegative cost c_e to each admissible probe. A minimum-total-cost complete bank is a minimum-weight basis of the observer matroid and can be found by the standard greedy basis algorithm.*

Exact cost optimality and robust design are different problems. For a basis J , let

$$G_J : K \rightarrow \mathbb{C}^d$$

be its response map and define

$$\gamma_J = \sigma_{\min}(G_J).$$

Definition 12.3 (Robust basis score). *For a fixed cardinality, energy, or cost class, the robust observer-design problem is*

$$\boxed{\max_J \gamma_J.}$$

Proposition 12.4 (Tight blind-sector optimum). *Suppose $r \geq d$ probe vectors on K have fixed total squared norm*

$$E = \sum_{a=1}^r \|g_a\|^2.$$

Let

$$S = \sum_{a=1}^r g_a^* g_a$$

be the blind-sector frame operator. Then

$$\lambda_{\min}(S) \leq \frac{E}{d},$$

with equality exactly when

$$S = \frac{E}{d} \text{Id}_K.$$

Thus tight banks maximize the worst blind-direction response at fixed total energy.

Proof. The average eigenvalue of S is

$$\frac{\text{tr } S}{d} = \frac{E}{d}.$$

The minimum eigenvalue cannot exceed the average, with equality exactly when all eigenvalues are equal. $\square$

Remark 12.5 (Incomplete catalogues). *If $\text{span } \mathcal{G} \neq K^*$, the same vector configuration still defines a representable matroid, but no subset of the catalogue is complete on K . Its matroid bases span only $\text{span } \mathcal{G}$, so they must not be called complete observer banks for the full blind space.*

Remark 12.6. *The matroid solves exact full reconstruction of the entire blind space. The singular-value score solves robustness. Greedy minimum-cost selection may therefore be followed by a conditioning refinement among equal-cost or near-minimal bases.*

For a proper target row space $T \subsetneq K^$, a fixed catalogue instead asks for a subset whose span contains T . As proved in Theorem 4.4, its minimal solutions need not be matroid bases and may require more than $\dim T$ catalogue probes.*

13 The degree-five 50/45/5 calibration

We now recover the exact finite model that motivated the abstract theory.

The companion spectral-blindness manuscript constructs a degree-five trace coordinate space X_5 with

$$\dim X_5 = 50.$$

The collapsed one-exponential-per-label observation has all-parameter rank

$$45,$$

hence blind kernel

$$K_5 = \ker A_{\text{coll}}, \quad \dim K_5 = 5.$$

Choose the ordered blind basis

$$(\eta_1, \eta_2, \beta_2, \beta_1, \kappa).$$

Let

$$\mathcal{G}_{\text{split}} : X_5 \longrightarrow \mathbb{C}^5$$

denote the ambient bank of five split-occurrence probes. Its restriction to the displayed blind basis is represented by

$$G_{\text{split}} = \begin{pmatrix} -6 & -6 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 & 0 \\ 0 & 0 & -12 & -6 & 0 \\ 0 & 0 & 6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 \end{pmatrix}.$$

Theorem 13.1 (Exact minimal lift of the degree-five blind sector). *The response matrix satisfies*

$$\det G_{\text{split}} = -2592 \neq 0.$$

Therefore the five split-occurrence probes form a complete observer basis on K_5 . Since $\dim K_5 = 5$, no bank of four scalar probes can repair the full collapsed map.

Proof. The nonzero determinant proves injectivity on K_5 . The lower bound on the number of scalar probes follows from rank-nullity. $\square$

The new theory adds conclusions not contained in the raw determinant calculation.

Corollary 13.2 (Finite parameter realization). *If the degree-five response family has polynomial degree at most m , then $m + 1$ distinct spectral values recover its exact all-parameter blind kernel. The resulting conditioning is controlled by the smallest singular value of the associated Vandermonde matrix.*

Corollary 13.3 (Robust repair constant). *Equip X_5 with a declared Hilbert structure and $\mathbb{C}^5$ with its Euclidean norm. If the displayed blind basis is chosen orthonormally in that structure, let σ be the lower bound of the collapsed observation on $K_5^\perp$, set*

$$\gamma = \sigma_{\min}(G_{\text{split}}),$$

and define the ambient cross-action bound

$$L = \|\mathcal{G}_{\text{split}}|_{K_5^\perp}\|.$$

Then the augmented observation

$$(A_{\text{coll}}, \mathcal{G}_{\text{split}})$$

is bounded below by

$$\delta(\sigma, \gamma, L) > 0$$

from Theorem 9.1, and deterministic noise ε produces reconstruction error at most ε/δ .

Remark 13.4 (Metric dependence of robustness). *The determinant and minimal channel count are coordinate-invariant. The singular value γ , the cross-action bound L , and the resulting noise constant depend on the declared Hilbert norms. They are therefore stated only after the metric has been fixed.*

Corollary 13.5 (Observer-matroid interpretation). *The five displayed probes form one basis of the observer matroid on K_5^* . Any other five admissible probes repair the blind sector exactly when their restriction matrix has rank five.*

Corollary 13.6 (Target-relative probe count). *Let $\Pi : X_5 \rightarrow Z$ be any linear target. The exact minimum number of supplemental scalar probes required beyond the collapsed observation is*

$$\boxed{\text{rank}(\Pi|_{K_5})}.$$

Thus five probes are necessary only when the target distinguishes all five blind modes. A target sensitive to a t -dimensional image of K_5 requires exactly t independent scalar probes.

Example 13.7 (Sector-specific repair). *The displayed response matrix is block triangular across the η -, β -, and κ -sectors. A target depending only on the single κ direction is repaired by the one probe whose response is 2κ . A target depending only on the two-dimensional η -sector requires exactly two independent probes. Full coordinate recovery requires all five.*

Corollary 13.8 (Response-adapted spectral sampling). *Let $\mathcal{R}_5$ be the coefficient-response space of the degree-five family. A sample set Q recovers its all-parameter blind kernel exactly when*

$$\mathcal{R}_5 \cap \ker(E_Q \otimes \text{Id}) = \{0\}.$$

Therefore the universal polynomial sample count is sufficient but need not be minimal for this particular response space.

Corollary 13.9 (Joint sample/probe profile). *For every spectral sample set Q , let A_Q be the sampled collapsed observation. The exact number of additional arbitrary scalar probes required for a target Π is*

$$\text{rank}(\Pi|_{\ker A_Q}).$$

Adding spectral samples lowers that count by exactly the target-recovery increment from Theorem 4.3.

Remark 13.10 (Information boundary). *The repair recovers the declared finite trace-coordinate model. It does not recover two different event histories that already produce the same endpoint operator. The trace-coordinate kernel and the endpoint-history kernel remain different objects.*

14 Degree-six exact stress test

The degree-five 50/45/5 model is small enough that its five blind modes can be displayed individually. A useful general theorem must also survive the next combinatorial layer without changing definitions. We therefore apply the same multiplicity, source-restriction, and split-occurrence constructions at total degree six.

For each multiplicity partition $\lambda \vdash 6$, let $\mathcal{N}_\lambda$ be the multiset cyclic necklaces and define the exact collapsed coefficient stack

$$\mathcal{M}_\lambda = \begin{pmatrix} B_\lambda D_1 P_\lambda \\ B_\lambda D_2 P_\lambda \\ \vdots \\ B_\lambda D_5 P_\lambda \end{pmatrix}.$$

All entries are integers. The rank computations below are exact over $\mathbb{Q}$, not floating-point estimates.

Theorem 14.1 (Exact degree-six collapsed classification). *The degree-six multiplicity partitions have the following necklace counts, collapsed all-parameter ranks, and blind dimensions.*

λ	necklaces	collapsed rank	blind dimension
(1, 1, 1, 1, 1, 1)	120	120	0
(2, 1, 1, 1, 1)	60	43	17
(2, 2, 1, 1)	30	13	17
(2, 2, 2)	16	5	11
(3, 1, 1, 1)	20	11	9
(3, 2, 1)	10	5	5
(3, 3)	4	3	1
(4, 1, 1)	5	4	1
(4, 2)	3	3	0
(5, 1)	1	1	0
(6)	1	1	0
<i>Total</i>	<i>270</i>	<i>209</i>	<i>61</i>

Hence the complete degree-six collapsed hierarchy has

270 formal necklace coordinates, 209 functional dimensions, 61 blind dimensions.

Proof. For every partition, occurrence-labelled cyclic classes are enumerated with one fixed cyclic anchor. They are collapsed to multiset necklaces by the incidence map P_λ . For each block source order and every cyclic-descent degree $1 \leq s \leq 5$, the integer rows of $B_\lambda D_s P_\lambda$ are formed exactly. Rational row reduction gives the displayed ranks. The accompanying verification script reproduces every entry and the three totals. $\square$

The large defect is not a failure of the split-occurrence representation. The leading-identity theorem still gives generic completeness of the rich split source module.

Theorem 14.2 (Exact minimal degree-six split lift at $q = 2$). *At $q = 2$, the split-occurrence response restricted to the exact collapsed blind kernel has full rank for every degree-six partition. The required ranks by partition are*

$$17, 17, 11, 9, 5, 1, 1$$

for

$$(2, 1, 1, 1, 1), (2, 2, 1, 1), (2, 2, 2), (3, 1, 1, 1), (3, 2, 1), (3, 3), (4, 1, 1),$$

respectively. Consequently there exists a bank of exactly

$$\boxed{61}$$

split-occurrence scalar probes that repairs the full degree-six collapsed map, and no smaller scalar bank can do so.

Proof. For each blind partition, let N_λ be an exact rational basis matrix for $\ker \mathcal{M}_\lambda$, and let $R_\lambda(2)$ be the full split-occurrence response matrix at $q = 2$. Exact row reduction gives

$$\text{rank}(R_\lambda(2)N_\lambda) = \dim \ker \mathcal{M}_\lambda$$

for every partition. Selecting pivot rows gives a split-probe bank with the stated cardinality. Summing the partitionwise ranks gives 61. Minimality follows from rank-nullity on the aggregate 61-dimensional blind space. $\square$

Corollary 14.3 (Target-specific degree-six count). *For any declared linear target $\Pi : X_6 \rightarrow Z$, the exact minimum number of arbitrary supplemental scalar probes is*

$$\boxed{\text{rank}(\Pi|_{K_6}), \quad \dim K_6 = 61.}$$

Thus the full 61-probe bank is necessary only for a target that distinguishes all degree-six blind modes.

Remark 14.4 (Growth of collapsed blindness). *The blind dimension rises from 5 at degree five to 61 at degree six. This does not indicate a failure of the general theory. It quantifies the increasing cost of observing repeated occurrences only through block source orders. The split-occurrence theorem identifies a complete source module, and the target-relative theory determines how much of the 61-dimensional defect a particular downstream conclusion actually needs to repair.*

15 Exact symmetry reduction of the degree-six blind sector

The 61-dimensional degree-six blind kernel carries natural symmetries. These are not additional assumptions. They arise because cyclic reversal and permutations of labels with equal multiplicity preserve both the multiset coordinate space and the collapsed source family.

For a partition λ , let $c_m(\lambda)$ denote the number of labels having multiplicity m . Define

$$\Gamma_\lambda = C_2^{\text{rev}} \times \prod_m S_{c_m(\lambda)}.$$

The first factor reverses cyclic orientation; each symmetric-group factor permutes labels having the same multiplicity. Cyclic orientation reversal here is a symmetry of trace words and is unrelated to spatial parity in the later operator application.

Theorem 15.1 (Degree-six equivariance). *For every $\lambda \vdash 6$, the collapsed coefficient stack $\mathcal{M}_\lambda$ intertwines the action of Γ_λ on necklace coordinates with the induced permutation of source-degree rows. Consequently*

$$K_\lambda = \ker \mathcal{M}_\lambda$$

is a Γ_λ -representation.

Proof. A multiplicity-preserving relabelling sends every block source to another block source and preserves relative cyclic descent. Simultaneous reversal of a source and a target also preserves relative cyclic descent. Target collapse commutes with both operations. Thus right action on necklace columns carries the row space of $\mathcal{M}_\lambda$ into itself, proving invariance of the kernel. $\square$

Let K_λ^+ and K_λ^- denote the cyclic-reversal even and odd subspaces. Exact rational character calculations give the following dimensions.

Theorem 15.2 (Exact reversal and invariant dimensions). *The nonzero degree-six blind sectors decompose as follows.*

λ	$\dim K_\lambda^+$	$\dim K_\lambda^-$	label invariant	fully invariant
(2, 1, 1, 1, 1)	5	12	0	0
(2, 2, 1, 1)	9	8	4	3
(2, 2, 2)	8	3	2	2
(3, 1, 1, 1)	5	4	1	0
(3, 2, 1)	3	2	5	3
(3, 3)	0	1	0	0
(4, 1, 1)	0	1	0	0
<i>Total</i>	30	31	12	8

Here “label invariant” means invariant under all permutations of labels with equal multiplicity, while “fully invariant” also imposes cyclic-reversal evenness. In particular,

$$\boxed{K_6 = K_6^+ \oplus K_6^-, \quad \dim K_6^+ = 30, \quad \dim K_6^- = 31,}$$

and the direct sum of fully invariant blind sectors has dimension eight.

Proof. For every group element, the verification script restricts its exact permutation matrix to an exact rational basis of K_λ . The traces are constant on conjugacy classes. Reversal-even and reversal-odd characters are obtained from

$$\chi_\pm(g) = \frac{1}{2}(\chi(g) \pm \chi(gJ)).$$

Character inner products give the isotypic multiplicities, while group averaging gives the invariant dimensions. Summing the exact partitionwise values yields 30, 31, 12, and 8. $\square$

For completeness, the nontrivial isotypic decompositions are:

λ	K_λ^+	K_λ^-
$(2, 1, 1, 1, 1)$	$[3, 1] \oplus [2, 2]$	$[3, 1] \oplus [2, 2] \oplus 2[2, 1, 1] \oplus [1, 1, 1, 1]$
$(2, 2, 2)$	$2\mathbf{1} \oplus 3\text{Std}$	$\text{sgn} \oplus \text{Std}$
$(3, 1, 1, 1)$	$\text{sgn} \oplus 2\text{Std}$	$\mathbf{1} \oplus \text{sgn} \oplus \text{Std}$

For $(2, 2, 1, 1)$, the label group is $C_2 \times C_2$. Ordering its four characters as $(++, +-, -+, --)$, the multiplicities are

$$(3, 1, 3, 2) \text{ on } K^+, \quad (1, 3, 2, 2) \text{ on } K^-.$$

The $(3, 2, 1)$ sector has trivial label symmetry and dimensions $(3, 2)$. The one-dimensional $(3, 3)$ and $(4, 1, 1)$ blind sectors are both reversal odd and antisymmetric under the available equal-multiplicity label swap.

Corollary 15.3 (Symmetry-reduced target counts). *Let Π be a degree-six target.*

(i) *If Π is cyclic-reversal even, then*

$$\mathfrak{B}_\Pi(A_{\text{coll}}) = \text{rank}(\Pi|_{K_6^+}) \leq 30.$$

(ii) *If Π is invariant under cyclic reversal and all label permutations preserving multiplicities, then*

$$\mathfrak{B}_\Pi(A_{\text{coll}}) \leq 8.$$

(iii) *Full coordinate recovery still has blindness index 61.*

Thus symmetry can reduce the exact arbitrary-probe requirement from 61 to at most 30, or to at most eight for a fully invariant target. These are target bounds, not claims that the omitted sectors cease to exist.

Theorem 15.4 (Symmetry-adapted degree-six split lift). *At $q = 2$, split-occurrence probes can be symmetrized and antisymmetrized under cyclic reversal so that*

$$30$$

even probe combinations form a basis on K_6^+ , and

$$31$$

odd probe combinations form a basis on K_6^- . Their union is a minimal 61-probe full lift.

Proof. The full split response is complete on K_6 and closed under reversal of the source word. Apply the dual idempotents $(I \pm J^*)/2$ to all split rows. Exact rational row reduction gives ranks 30 and 31 on the corresponding blind subspaces. The verification output records explicit source/reversed-source pairs for both selected banks. Minimality follows from the two parity dimensions. $\square$

The symmetry ledger is useful only after the downstream target declares its transformation law. A reversal-invariant scalar cannot detect reversal-odd modes, while an orientation-sensitive target may require them. The target map, not aesthetic preference, decides which symmetry sectors are lawful to discard.

16 Application to finite operator cuts

Let W be a bounded self-adjoint operator on a Hilbert space X , and let

$$X = V \oplus V^\perp$$

be a finite cut. Write

$$W = \begin{pmatrix} A & C^* \\ C & D \end{pmatrix}.$$

A positive finite block $A > 0$ does not determine the sign of W : the omitted complement is a target-relevant blind sector for the global positivity predicate.

The exact Feshbach complement is

$$F = D - CA^{-1}C^*.$$

Proposition 16.1 (Finite block plus lawful complement). *If $A > 0$, then*

$$W \geq 0 \iff F \geq 0.$$

Thus F is the exact supplemental lift of the finite block for the global sign target.

Proof. Use the triangular congruence

$$\begin{pmatrix} I & 0 \\ -CA^{-1} & I \end{pmatrix} W \begin{pmatrix} I & -A^{-1}C^* \\ 0 & I \end{pmatrix} = \begin{pmatrix} A & 0 \\ 0 & F \end{pmatrix}.$$

$\square$

This fits the blindness exact sequence. The finite compression observes the V -block; the second-stage global target is blind to any negative direction inside $V^\perp$ unless the complement is consumed.

For compact relative operators, a different but related lift is available. Suppose

$$W = R^{1/2}(I - K)R^{1/2},$$

where $R > 0$ and K is compact self-adjoint. Let $P_n \rightarrow I$ strongly be finite-rank orthogonal projections. Then

$$\|K - P_n K P_n\| \rightarrow 0.$$

Writing

$$\mu_n = \sup \sigma(P_n K P_n|_{P_n X}),$$

one has

$$\sup \sigma(K) \leq \mu_n + \|K - P_n K P_n\|.$$

Corollary 16.2 (Matched finite-section certificate). *If*

$$\mu_n + \|K - P_n K P_n\| \leq 1,$$

then $W \geq 0$.

Remark 16.3 (Why the restricted spectrum is declared). *Viewed on the whole space, $P_n K P_n$ has an artificial zero eigenspace on $(I - P_n)X$. The finite extremal quantity is therefore the spectral supremum of the compression on $P_n X$, not an ambiguous largest eigenvalue of the zero-extended operator.*

Remark 16.4 (Projection consistency). *The finite extremal eigenvalue and the off-space tail must refer to the same projection P_n . A coefficient tail for one approximation cannot repair blind directions omitted by an unrelated finite source space.*

The operator application illustrates the main principle of the paper:

A finite packet becomes globally meaningful only after its target-relevant blind complement is represented by a lawful lift with a quantitative margin that survives completion.

17 Parity–Feshbach carrier blindness

The target-blindness calculus also applies to finite carrier modules arising from a parity decomposition followed by a Feshbach reduction. The application is informational rather than a sign proof: positivity is nonlinear, whereas the blindness index is defined for linear carrier targets.

Let a finite carrier cut be

$$H_N = H_N^+ \oplus E \oplus Q_N^-,$$

where E is a fixed finite odd block, H_N^+ is a finite even carrier, and Q_N^- is a finite odd-complement carrier. Put

$$\dim H_N^+ = n_+, \quad \dim Q_N^- = n_-.$$

Let P_+ , P_E , and P_F be the three coordinate projections.

Theorem 17.1 (Finite parity–Feshbach carrier ledger). *For the finite packet $A_{E,N} = P_E$,*

$$\mathfrak{B}_{P_+}(A_{E,N}) = n_+, \quad \mathfrak{B}_{P_F}(A_{E,N}) = n_-,$$

and

$$\mathfrak{B}_{I_{H_N}}(A_{E,N}) = n_+ + n_-.$$

After adding the odd-complement carrier,

$$A_{EF,N} = (P_E, P_F),$$

one has

$$\mathfrak{B}_{P_F}(A_{EF,N}) = 0, \quad \mathfrak{B}_{P_+}(A_{EF,N}) = \mathfrak{B}_{I_{H_N}}(A_{EF,N}) = n_+.$$

After also adding the even carrier, the observation

$$(P_E, P_F, P_+)$$

is injective and every displayed carrier blindness index is zero.

Proof. The three kernels are, respectively,

$$H_N^+ \oplus Q_N^-, \quad H_N^+, \quad \{0\}.$$

Restricting the declared target projections to those kernels gives the stated ranks. $\square$

The exact target-blindness increments are therefore

$$(n_+ + n_-) - n_+ = n_-$$

for the odd-complement lift and

$$n_+ - 0 = n_+$$

for the even lift. The two obligations are disjoint. Strengthening a finite odd packet cannot reduce the even-carrier index.

In an operator realization, the odd complement is represented by a Feshbach operator F^- , while the independent even carrier is an operator W^+ . These are operator-valued lifts, not single scalar probes. If $n_+(N), n_-(N) \rightarrow \infty$, then the full-carrier blindness of a fixed finite packet grows like

$$n_+(N) + n_-(N).$$

A fixed finite scalar bank cannot repair this growing carrier. Completion requires operator-valued carriers, growing finite banks with uniform lower margins, or a theorem proving the omitted carrier target-null.

Remark 17.2 (Sign boundary). *Carrier completeness does not imply positivity. Once both carriers are present, the remaining terminal obligations are still independent spectral inequalities for the even operator and the odd Feshbach complement. The blindness ledger prevents an unobserved sector from being silently dropped; it does not replace the no-crossing theorem that decides sign.*

18 Boundaries and future directions

The paper proves a layered reconstruction package:

- finite and response-adapted parameter sampling
- + joint sample/probe design
- + ordinary, target-valued, and equivariant blindness composition
- + full and target-relative minimal lifts
- + robust decoders and a posteriori response certificates
- + completion-stable reconstruction.

The following statements are proved within the declared scope:

- unisolvent samples recover the full coefficient-stack kernel, while a smaller response-adapted set suffices exactly when evaluation is injective on the actual coefficient-response space;
- after fixing samples Q , the exact minimum number of arbitrary scalar probes is

$$\text{rank}(\Pi|_{\ker \mathcal{E}_Q});$$

- fixed-catalogue target completion is a prescribed-subspace spanning problem and need not be a matroid-basis problem;
- ordinary blindness and target blindness obey exact stage-by-stage composition laws;
- symmetry types absent from the target codomain are target-null, and invariant targets see only invariant blind sectors;
- target faithfulness admits a bounded decoder exactly when the target stability modulus is positive;

- approximate response lifts admit an a posteriori defect bound in the source metric, and response–Krylov enrichment leaves only the newest frontier uncaptured;
- a finite compression, its cross action, the complement spectral bound, and the operator tail combine through one matched two-block certificate;
- full and target-relative repaired observations survive strong completion under uniform hypotheses;
- the degree-five 50/45/5 model admits an exact minimal five-probe full lift;
- the degree-six hierarchy has 270 formal coordinates, rank 209, blind dimension 61, and an exact minimal 61-probe split lift at $q = 2$;
- the degree-six blind kernel decomposes into 30 cyclic-reversal-even and 31 cyclic-reversal-odd dimensions, with a fully invariant sector of dimension eight;
- finite parity–Feshbach cuts have an exact carrier-blindness ledger that separates the even and odd-complement obligations.

The following remain outside the present theorem:

- nonlinear global observation maps without uniform metric regularity;
- unbounded operators with changing domains, except after a separate graph, form, or resolvent transport theorem;
- sharp probabilistic sample/probe design under correlated noise;
- a general efficient algorithm for minimum-cost target completion from an arbitrary fixed probe catalogue;
- endpoint-history recovery after exact endpoint collapse;
- dimension-specific matrix trace identities beyond the declared formal coordinate model;
- promotion from carrier completeness to operator positivity;
- any global sign theorem or open conjecture.

Natural next developments include unbounded graph-, form-, and resolvent-stable target lifts; robust symmetry-constrained catalogue design; and uniform response-residual margins for varying finite-section spaces.

The degree-six symmetry result shows why target-relative recovery is essential: full collapsed reconstruction costs 61 scalar lifts, a reversal-even target can see at most 30 blind dimensions, and a fully invariant target can see at most eight. The target transformation law, not the size of the native blind kernel alone, determines the lawful repair cost.

References

- [1] M. Dabas, Geometric Completion of Thermodynamic Response: From the T – V – S – P Compass to Recursive λ -Jacobian Geometry, *arXiv preprint* arXiv:2603.20773, 2026. doi:10.48550/arXiv.2603.20773.
- [2] M. Dabas, When Spectra Forget Order: Spectral Blindness, Observer-Marked Separation, and Stable Reconstruction of Ordered Matrix Products, companion preprint, version 0.9, 2026.

- [3] B. Adcock and A. C. Hansen, A generalized sampling theorem for stable reconstructions in arbitrary bases, *Journal of Fourier Analysis and Applications* 18(4) (2012), 685–716. doi:10.1007/s00041-012-9221-x.
- [4] B. Adcock, A. C. Hansen, E. Herrholz, and G. Teschke, Generalized sampling: extension to frames and inverse and ill-posed problems, *Inverse Problems* 29(1) (2013), 015008. doi:10.1088/0266-5611/29/1/015008.
- [5] M. Fickus, D. G. Mixon, and M. J. Poteet, Frame completions for optimally robust reconstruction, *Proceedings of SPIE* 8138 (2011), 81380Q, 1–8. arXiv:1107.1912.
- [6] D. R. Larson and S. L. Scholze, Signal reconstruction from frame and sampling erasures, *Journal of Fourier Analysis and Applications* 21(5) (2015), 1146–1167. doi:10.1007/s00041-015-9404-3.
- [7] P. Zhu, M. E. Argentati, and A. V. Knyazev, Bounds for the Rayleigh quotient and the spectrum of self-adjoint operators, *SIAM Journal on Matrix Analysis and Applications* 34(1) (2013), 244–256. doi:10.1137/120884468.
- [8] R. G. Douglas, On majorization, factorization, and range inclusion of operators on Hilbert space, *Proceedings of the American Mathematical Society* 17(2) (1966), 413–415. doi:10.1090/S0002-9939-1966-0203464-1.
- [9] R. A. Horn and C. R. Johnson, *Matrix Analysis*, second edition, Cambridge University Press, 2013.
- [10] G. H. Golub and C. F. Van Loan, *Matrix Computations*, fourth edition, Johns Hopkins University Press, 2013.
- [11] R. J. Duffin and A. C. Schaeffer, A class of nonharmonic Fourier series, *Transactions of the American Mathematical Society* 72 (1952), 341–366.
- [12] O. Christensen, *An Introduction to Frames and Riesz Bases*, second edition, Birkhäuser, 2016.
- [13] T. Kato, *Perturbation Theory for Linear Operators*, Springer, 1995.
- [14] J. Oxley, *Matroid Theory*, second edition, Oxford University Press, 2011.